

HIDDEN COSTS OF PIGGERY OPERATIONS IN INDIA

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*How India's Unregulated Piggery
Sector Threatens Public Health*

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Introduction

1.1. A Bird's Eye View - Piggery Operations in India

Piggery units are a rapidly growing yet overlooked and unregulated part of India's agricultural landscape, particularly in rural regions, where their impact on local economies and food supply varies. The role of such units within the broader agricultural landscape of India reflects a complex interplay of economic, social, and environmental factors that shape their significance in different contexts across the country.

Historically, piggery operations have been prevalent in north-eastern states such as Assam, Nagaland, and Manipur, where pork is a dietary staple. The Indian government has implemented various schemes to support this sector, including financial assistance, with the intent to enhance rural livelihoods.¹ However, the rapid and unregulated expansion of piggery operations raises critical concerns regarding environmental sustainability and public health.

According to the 20th Livestock Census conducted by the Department of Animal Husbandry and Dairying, Ministry of Fisheries, Animal Husbandry, and Dairying, Government of India in 2019, the population of pigs in India was approximately 9.06 million,² with significant concentrations in states like Uttar Pradesh, West Bengal, and Assam. The industry is characterized by a wide array of production systems ranging from small-scale backyard units to larger commercial operations. Across the different types of production systems, this sector faces substantial challenges such as inadequate infrastructure for waste management.³ Such conditions compromise animal welfare and give rise to recurring outbreaks for highly virulent diseases like African Swine Fever.⁴



1.2. Overview of the Piggery Industry in India

While India has a substantial pig population, the industry is often characterized by its challenges rather than its successes. The industry's growth has raised critical questions about its sustainability and ethical implications, especially regarding animal welfare and its detrimental environmental impact.

1.2.1. *Economic Significance and Market Dynamics*

Piggery operations are often portrayed as a viable economic activity for many rural households, providing a source of income and nutrition. It is generally held that pigs have a high Feed Conversion Ratio (FCR) which is a measure of how effectively an animal turns input feed into its body weight as output. However, in field settings there are multiple factors affecting the daily feed intake and FCR of pigs.⁵ Research has shown that pig behaviour is one such factor that also influences FCR,⁶ indicating a need to focus on the health, temperament and engagement/enrichment for pigs, to help improve the behaviours of pigs for a better FCR.

Despite the perceived economic advantages, the industry faces numerous challenges that hinder its growth and sustainability. While larger commercial operations may benefit from economies of scale, they frequently do so at the expense of animal welfare, sustainability and public health. The focus on maximizing production efficiency can lead to overcrowding and inadequate living conditions for pigs, raising serious ethical concerns, as well as rapid environmental degradation of surrounding areas.

Might a transition to more environmentally sustainable piggery operations not offer the operators enhanced long-term stability and ethical accountability, while simultaneously safeguarding and prioritising public health?



1.2.2. *Production Systems*

The production systems in the Indian piggery sector can be broadly categorized into three types:

TRADITIONAL BACKYARD SYSTEMS:

These systems involve small numbers of pigs, typically up to 10 in number. Pigs in this system are raised primarily for household consumption or supply of pork to local markets. This system generally entails minimal investment, it tends to be characterised by inadequate housing. The reliance on scavenging for food by pigs in these systems can lead to nutritional deficiencies and increased susceptibility to diseases.⁷

SEMI-INTENSIVE SYSTEMS:

These systems typically involve medium herd sizes with 50-200 pigs.⁸ Piggery operators create enclosures where pigs are reared. These systems have increased control over pigs than traditional systems, but the concentration of pigs at single premises warrants more attention to waste management infrastructure and disease control.

INTENSIVE SYSTEMS:

These operations with more than 200 pigs focus on maximizing production productivity.⁹ These systems with high stocking density require high degree of attention to waste management including the mitigation of gaseous emissions that arise from crowded premises with humidity and gas build up.¹⁰

1.2.3. *Challenges Arising from Poor Management of Piggery Operations*

The piggery industry's growing emphasis on economic growth poses the potential for overlooking the ethical implications, particularly concerning animal welfare, environmental waste and public health. The Indian piggery industry is fraught with numerous challenges due to poor management protocol and lack of regulatory oversight:

Waste Management & Public Health Compromise: Proper waste management is critical for both environmental protection and animal welfare. Most piggery units lack adequate infrastructure for managing solid and liquid waste. Wastewater is often discharged into drains or fields untreated, causing soil and water degradation and pollution. Unmitigated waste breeds diseases that can affect public health.

Disease Outbreaks: The piggery industry is vulnerable to various diseases that can devastate herds and impact food security. Diseases such as African Swine Fever (ASF)¹¹ have emerged as significant threats, leading to mass culling and economic losses. The lack of biosecurity measures on many units exacerbates these risks.

Animal Welfare Concerns: The United Kingdom's Farm Animal Welfare Council codified the Five Freedoms of Animal Welfare in 1979, endorsing them as "the best possible standards of welfare for animals." These are a set of principles that guide the humane treatment of animals and include: a) Freedom from hunger and thirst; b) Freedom from discomfort; c) Freedom from pain, injury, or disease; d) Freedom to express normal behaviour; and e) Freedom from fear and distress. These freedoms serve as a foundation for ensuring the ethical care and well-being of animals globally. Intensive pig husbandry can lead to poor living conditions for pigs, including overcrowding and lack of access to outdoor spaces. A vast majority of piggery establishments still have much to do to meet standards of the Five Freedoms, often referred to as the "Five Freedoms of Animal Welfare".



Purpose & Scope

2.1. Objectives of the Research

Through this research, an attempt is made to examine the status of waste management by piggery operations in India and implications of the same for public health. The paper seeks to explore the prevalent state of waste management at piggeries across India – focusing on solid, liquid and biomedical waste management. The impact of such practices on public health are also examined. Attention is also paid to siting parameters followed by piggery operators and the consequences for the same. The paper identifies specific public health risks and the propensity for non-regulation of piggery operations to give rise to specific diseases. The paper concludes by identifying potential measures that can be employed to address the same.

The significance of this research draws from three key research constructs:

A What kind of solid, liquid and gaseous waste is generated by piggery establishments and what is the pollution potential arising thereby?

Piggery operations generate significant waste, including dung, carcasses, leftover feed, and cleaning wastewater. This paper analyzes the state of pollution on the ground, the waste discarded, current management methods, and the resulting impact on soil, water, and air.

B. How does inadequate waste management by piggery establishments affect public health and safety?

The paper explores what the consequences of poor waste management on public health and safety may manifest as. Specifically, the incidence of diseases that may arise from mismanagement of piggery waste is detailed. These comprise direct pathways through which piggery waste management ties to human health.

C. What indirect risks are associated with waste management at piggery establishments?

The paper delves into the relationship between pigs being reared in high density environments which are unhygienic and the heightened potential for 6 non-therapeutic antibiotic usage. Indiscriminate antibiotic usage may give rise to antimicrobial resistance (AMR).

Specifics of Observed Waste from Piggery Units

The management of piggery waste has emerged as a critical issue in India, where the rapid intensification of piggery operations has raised significant environmental and public health concerns. This section seeks to analyse findings from a study conducted among piggery units in India, aimed at examining various aspects of waste management practices followed in the piggeries across the country, focusing on the implications of these practices for local ecosystems and communities. Through the comprehensive study, data was collected on several key factors related to waste disposal methods and their associated risks.

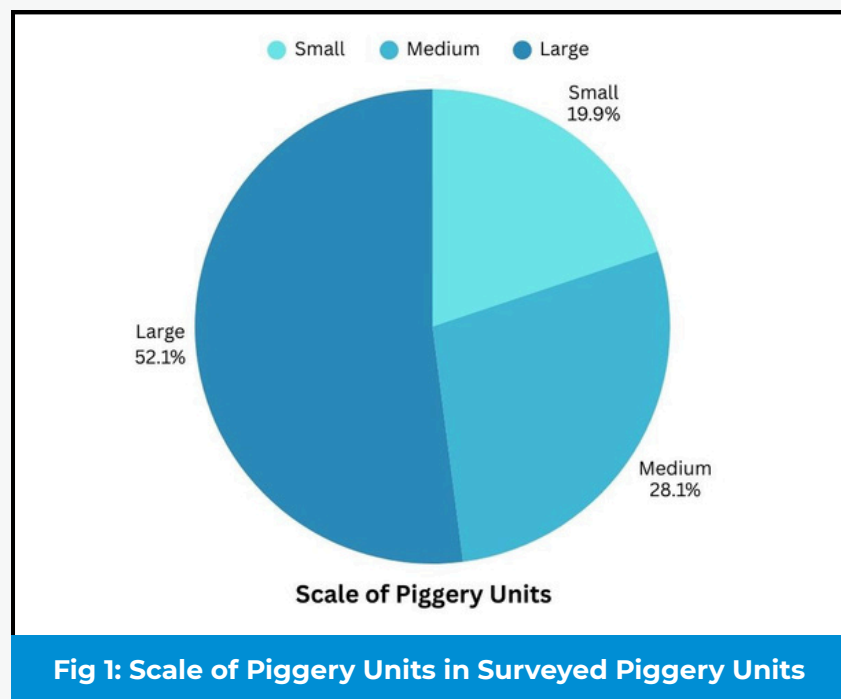
The study addressed critical issues such as the disposal of untreated wastewater and slurry, which can lead to severe environmental pollution when discarded in drains or agricultural fields. Additionally, the presence of slurry pits was examined as they may create breeding grounds for disease vectors, particularly mosquitoes, which pose risks for vector-borne diseases. The absence of infrastructure for treating solid waste was also assessed, alongside practices related to the disposal of solid waste in water bodies and drains.

Furthermore, the study evaluated the mechanisms in place for managing biomedical waste generated from veterinary practices associated with piggery operations. The potential for fly infestations and odour nuisances due to inadequate waste management was also documented, as these issues can significantly affect the quality of life for nearby residents. The study also considered carcass disposal methods and whether siting parameters met environmental and health standards.

By analysing these factors, this study aims to provide a clearer understanding of the current state of piggery waste management in India and its implications for public health and environmental sustainability. The findings will contribute to ongoing discussions about improving waste management practices in the livestock sector, ultimately promoting better health outcomes for communities and protecting local ecosystems.

3.1. Study Overview

The study includes a diverse range of piggery units categorized by size: small (up to 49 pigs), medium (50-199 pigs), and large (over 200 pigs). Data was collected on various waste management practices, with responses for specific questions (explained as “headers”) recorded as “Yes” (the practice is employed), “No” (the practice is not employed), or “NA” (Not Applicable). The inclusion of the “NA” category is crucial, as it accounts for instances where a specific waste management practice or parameter could not be observed or did not apply to a particular piggery unit due to its scope of operations or setup. The study encompassed 146 piggery units across eight states, including Delhi, Uttar Pradesh, Haryana, Rajasthan, Karnataka, Madhya Pradesh, Nagaland, and Assam.



3.2. Explanation of Headers of Survey

The following headers will guide the discussion, providing a comprehensive overview of the current practices and their implications.

Untreated Wastewater Discarded in Drains: This header assesses whether untreated wastewater generated from piggery operations was improperly disposed of into public drainage systems, which could result in significant environmental pollution and public health risks due to the introduction of pathogens and nutrients into urban waterways. Faecal contamination of public drains leads to several invasive and antibiotic resistant organisms such as *Salmonella enterica serovar Typhimurium*. This pathogen, a known cause of Typhoid fever, poses a significant threat to human health.¹²

Untreated Wastewater/Slurry Discarded in Agricultural Fields: This header evaluates if piggery units released untreated wastewater or slurry directly onto agricultural fields, a practice that could lead to soil and water contamination, nutrient overloads, and potential health hazards for consumers due to the presence of pathogens and heavy metals in the waste.¹³

Pits with Slurry/Effluent Creating a Breeding Ground for Vector-Borne Diseases: This header analyses whether the presence of open pits containing slurry or effluent contributed to the proliferation of disease-carrying vectors, such as mosquitoes, which could have led to increased incidences of vector-borne diseases like dengue fever, West Nile virus or Japanese Encephalitis in surrounding communities.

Absence of Infrastructure to Treat Solid Waste: This header examines whether piggery operations lacked adequate facilities or systems for the proper treatment and disposal of solid waste, which could have led to improper disposal practices that exacerbated environmental degradation and public health concerns.

Solid Waste Discarded in Water Bodies: This header looks into whether solid waste from piggery operations was unscientifically disposed of in nearby rivers, lakes, or ponds, posing serious risks to aquatic ecosystems and potentially contaminating drinking water sources for local communities.

Solid Waste Discarded in Drains: This header assesses if solid waste generated from piggery operations was improperly thrown into drainage systems, which could have contributed to blockages, flooding, and the spread of pollutants that affected both urban infrastructure and public health.

Absence of Mechanism to Segregate and Safely Discard Biomedical Waste: This header evaluates whether there were established protocols or systems for the segregation and safe disposal of biomedical waste associated with piggery operations, which is crucial for preventing contamination, managing diseases in pigs and protecting public health. Improper disposal of medical waste, including needles and pharmaceutical residues could increase the risk of zoonotic and other microbial diseases by contaminating ecosystems and facilitating pathogen transmission. Such practices pose serious threats to both human and environmental health, heighten the spread of antimicrobial resistance¹⁴ and underscore the need for stringent waste management protocols. Improper disposal of medical waste such as needles and antibiotic residues could increase the risk of antimicrobial resistance (AMR), which is now classified by WHO as one of the top global health threats.

Fly Infestation & Odour Nuisance Owing to Swill and Other Reasons: This header investigates whether inadequate management of organic waste, particularly swill feed and other by-products, led to fly infestations and persistent odour problems that affected the quality of life for nearby residents. Flies are known mechanical vectors of pathogens including Salmonella, E. coli, and parasitic eggs,¹⁵ increasing the risk for malaria, typhoid fever, cholera, dysentery, and food poisoning.

Inappropriate Carcass Disposal: This header assesses whether piggery units employed improper methods for disposing of pig carcasses, which could have posed significant biosecurity risks, attracted scavengers, and contributed to the spread of zoonotic disease among livestock and humans alike. 10. Inappropriate

Siting Parameters: This header evaluates whether the locations of the piggery operations adhered to appropriate environmental and health standards regarding waste management practices; improper siting could have led to increased pollution risks and negative impacts on surrounding ecosystems and communities.

3.2. Key Findings

UNTREATED WASTEWATER DISPOSAL:

A significant number of piggery units reported discarding untreated wastewater either into drains or in agricultural fields or in open pits/ponds. This practice raises concerns about environmental pollution and potential contamination of water sources. The high prevalence of untreated wastewater disposal indicates a lack of effective treatment facilities.

- Of the 146 units visited, methodology for wastewater disposal could not be observed or ascertained at 4 units.
- Out of the remaining 142 piggery units surveyed during the study, a significant proportion, specifically 83 units, were found to be discharging untreated wastewater directly into public drainage systems.
- Additionally, 74 were observed to be disposing of untreated wastewater or slurry directly onto agricultural fields.

- Further, 42 units were found to be discarding their waste into open pits containing slurry or effluent, resulting in stagnant accumulations of organic waste, creating conditions highly conducive to the proliferation of disease vectors, including mosquitoes and flies, thereby posing a significant risk for pathogenic and microbial infections, which poses health hazards.
- Of the 142 piggery units where wastewater disposal could be observed/ascertained, merely 5 units were found to be fully compliant with the limited waste management protocols. These compliant units ensured that:
 - Untreated wastewater was not discharged into public drains
 - Slurry or wastewater was not applied to agricultural land
 - No open pits containing slurry or effluent were present

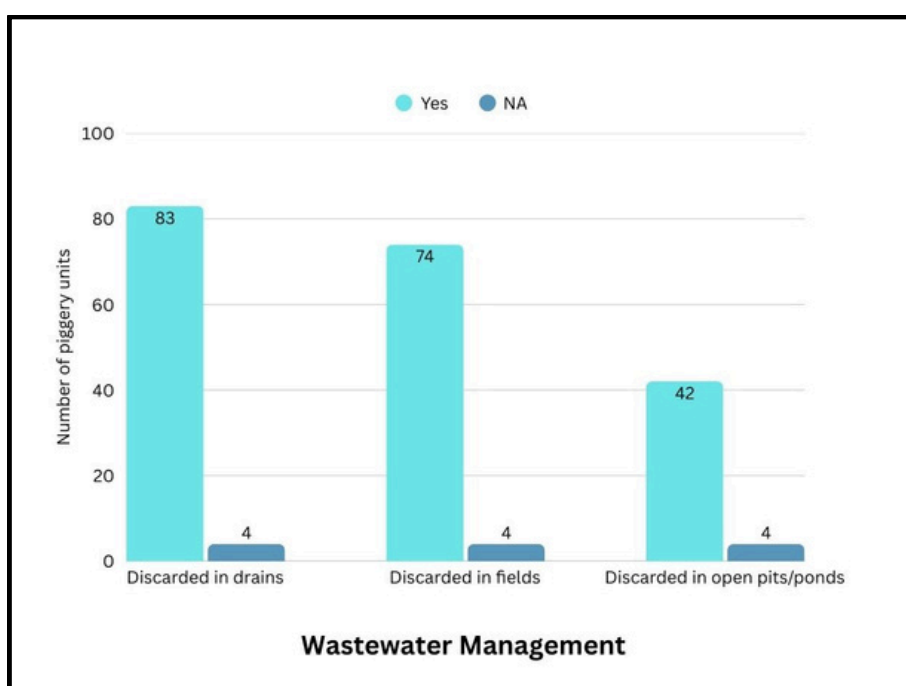


Fig 2. Wastewater Management: Over half of observable units (83/142) discard untreated wastewater into public drains, posing a critical public health risk.

SOLID WASTE MANAGEMENT:

- Of the 146 piggery units surveyed, a staggering number of 144 units were found to lack the necessary infrastructure for the treatment of solid waste.
- Further, 31 units were found to be discarding solid waste directly into water bodies.
- Additionally, 81 were found to be discarding solid waste into public or municipal drains, while in 4 units it was not apparent that their waste disposal practices differed from those observed in the surveyed units.

- Of the 146 piggery units surveyed, not a single one was found to be fully compliant in the areas of solid waste management. Specifically, all units lacked the necessary infrastructure to treat solid waste, and significant numbers were found to be discarding solid waste into water bodies and public drains.



BIOMEDICAL WASTE MANAGEMENT:

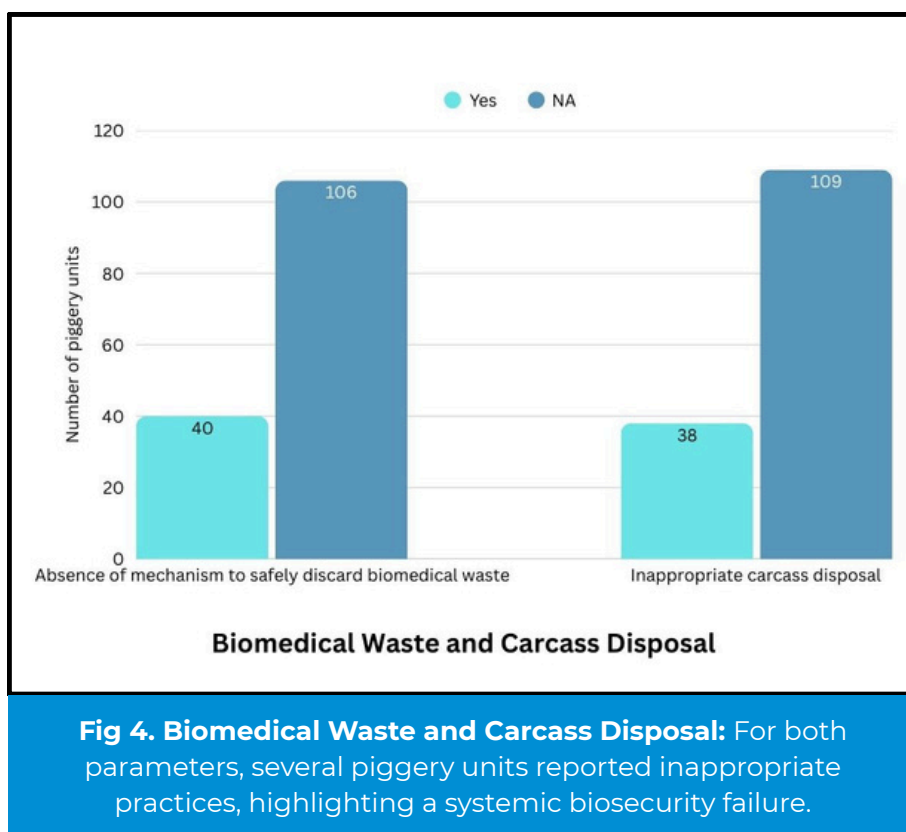
During the study conducted on ground, the methodology employed for biomedical waste management could be observed at 40 out of 146 units. All 40 units were found to lack proper waste segregation systems, highlighting a widespread and systemic failure to implement even basic biomedical waste disposal protocols.

ODOUR AND FLY INFESTATION:

Assessment of odour and fly infestation was not possible at 1 out of the 146 piggery units surveyed. A staggering 145 piggery units were found to have severe fly infestations and odour nuisances caused by swill and other poor waste management systems.

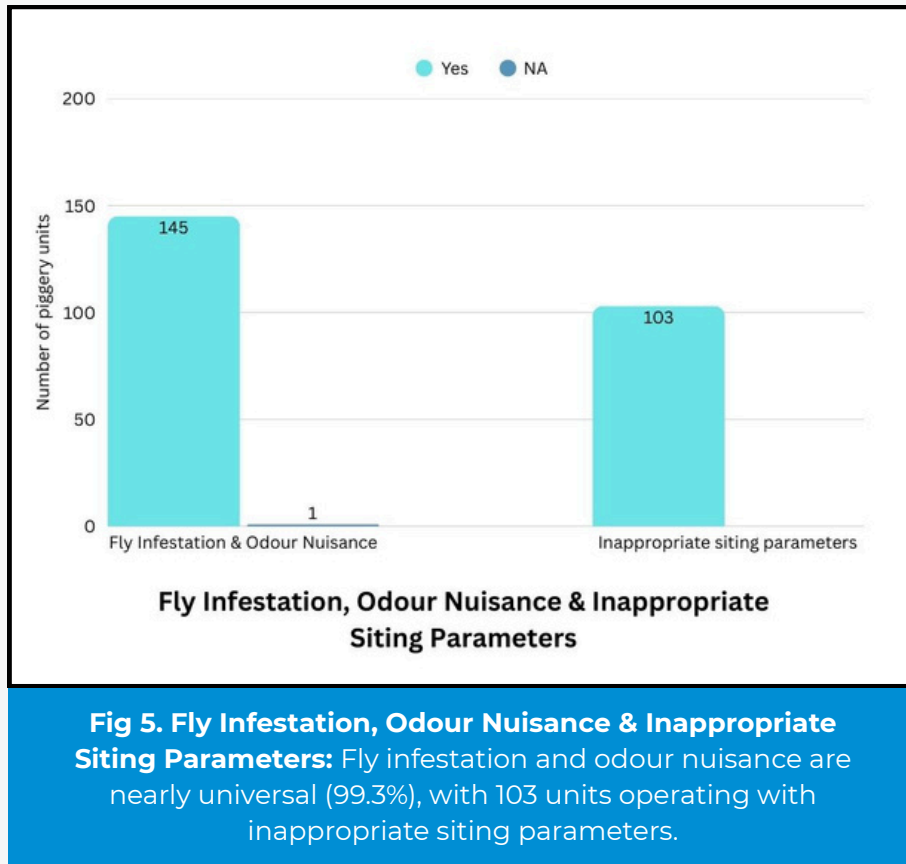
INAPPROPRIATE CARCASS DISPOSAL:

During the study conducted on ground, the methodology employed for carcass collection and disposal could be observed at 38 out of 146 units. Each of these 38 units were found to employ inappropriate carcass disposal practices, raising serious biosecurity concerns and increasing the risk of disease transmission and environmental contamination. The accumulation of organic waste, including carcasses and effluent byproducts, often exceeds the absorptive capacity of the surrounding environment, leading to risks such as leachate formation, nitrate infiltration, and habitat disruption. This reflects a systemic failure to implement even the most basic sanitary and biosecurity protocols, highlighting a disturbing lack of accountability and a blatant disregard for both animal welfare and environmental health standards.



SITING PARAMETERS:

Of the 146 piggery units surveyed, 103 were found to be operating with inappropriate siting parameters. This included piggery units being established and operated in close proximity to residential areas, schools, colleges, hospitals, piggeries/other livestock units and National and State Highways.



Nature & Impact of Waste Discarded from Piggery Operations

Waste generated from piggery units consists of both solid and liquid waste. This waste includes faeces, urine, and leftover feed. The composition and characteristics of piggery waste are crucial for understanding its environmental impact and potential uses in agriculture.

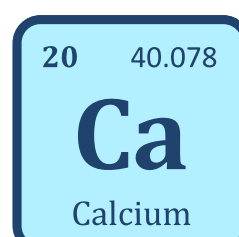
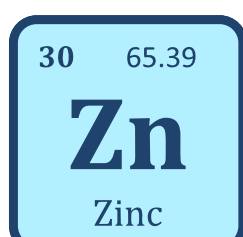
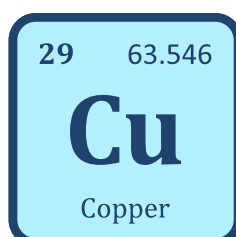
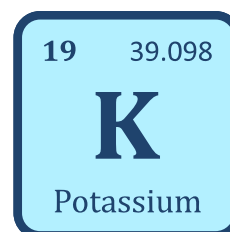
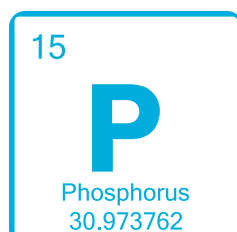


4.1. Composition of Piggery Waste

Nutrients: Piggery waste is rich in essential nutrients, particularly nitrogen (N), phosphorus (P), and potassium (K). The nutrient composition varies based on factors such as the type of pigs, their diet, and the management practices employed on the unit. For instance, nitrogen content can range from 0.20% in piglets to 0.57% in finishers, while phosphorus content increases from 0.02% to 0.12% as pigs mature.¹⁶ The organic matter content typically constitutes about 70-80% of the dry matter in pig slurry, which averages around 6-8% dry matter.¹⁷

Pathogens: Piggery waste can serve as reservoir for various pathogens, including bacteria such as Salmonella and Escherichia coli,¹⁸ which pose health risks to humans and animals. The presence of these pathogens is exacerbated by the storage methods used in intensive piggery operations, such as lagoons or pits that may go untreated before being released into the environment. The spread of these pathogens can lead to significant public health concerns, particularly in communities near piggery units, such as mucosal irritation,¹⁹ respiratory ailment,²⁰ and an overall decreased quality of life.²¹

Heavy Metals: Heavy metals such as copper (Cu), zinc (Zn), lead (Pb), and cadmium (Cd) are commonly found in pig waste due to their accumulation through feed and environmental exposure.²² Excessive levels of heavy metals can be toxic and have detrimental effects on both animal and human health.²³ For instance, lead and cadmium exposure has been linked to kidney damage and other health issues in livestock.²⁴ Studies have also shown that heavy metal concentrations in pig manure can exceed safe limits for agricultural use, raising concerns about soil contamination and food safety.²⁵



4.2. Physicochemical Properties of Piggery Wastes

The physicochemical properties of piggery waste are critical for understanding its behaviour during storage, treatment, and application in agriculture. These properties include pH levels, organic matter content, water content, total solids, and volatile fatty acids.

pH Levels: The pH of piggery waste is typically alkaline, ranging from approximately 6.5 to 9.5, depending on the age and composition of the waste. Fresh pig excreta generally has a lower pH, however, but as the waste accumulates, the buildup of ammonia and other alkaline compounds increases the pH levels, exacerbating its environmental nuisance.²⁶ For instance, studies have reported pH values ranging from 7.66 to 8.7 in various pig slurry samples.²⁷ The alkalinity is influenced by the feed composition and the microbial activity during anaerobic digestion processes.

Organic Matter: Content Piggery waste is composed predominantly of organic matter, which can account for about 70-80% of its dry weight.²⁸ This organic matter includes undigested feed, faecal material, and bedding materials. The high organic content facilitates microbial activity during anaerobic digestion, which is essential for biogas production. Studies indicate that piggery waste has a high concentration of volatile solids (VS),²⁹ often exceeding 70% of total solids.³⁰ The presence of complex organic molecules in the waste also affects its biodegradability and nutrient release patterns.

Water Content: Piggery waste is characterized by high moisture content, typically between 80-90%.³¹ This high-water content affects the handling and storage of the waste, making it more challenging to manage compared to drier wastes. At lower concentrations, piggery waste moves freely, but as it becomes more concentrated, it forms a dense slurry, posing challenges for waste management and treatment systems.³²

Volatile Fatty Acids (VFAs): The presence of Volatile Fatty Acids (VFAs) is critical for determining the stability and efficiency of anaerobic digestion of piggery waste. High VFA concentrations signal incomplete digestion and process instability, often leading to operational challenges like acidification. This instability compromises treatment system efficiency and increases the risk of odor and microbial emissions.³³ Typical VFA concentrations in pig slurry can exceed 10,000 mg/L under certain conditions.³⁴

Nutrient Composition: Piggery waste is a complex mixture that not only contains heavy metals and pathogens but is also rich in essential nutrients such as nitrogen (N), phosphorus (P), and potassium (K). While these nutrients are vital for plant growth, their presence in untreated piggery waste can lead to significant environmental and animal welfare concerns.

- **Nitrogen (N):** Excessive nitrogen in piggery waste can have detrimental effects on both the environment and animal health. When untreated waste is applied to agricultural land, nitrogen can leach into groundwater, leading to contamination of drinking water supplies. High levels of nitrates in water pose serious health risks, particularly for vulnerable populations such as infants, causing conditions like methemoglobinemia or "blue baby syndrome."³⁵ Furthermore, the accumulation of nitrogen in soils can disrupt local ecosystems and contribute to soil degradation.³⁶
- **Phosphorus (P):** The high phosphorus content in untreated piggery waste can lead to severe environmental issues when it runs off into water bodies. This runoff contributes to eutrophication, a process that causes harmful algal blooms.³⁷ These blooms deplete oxygen levels in the water, resulting in dead zones where aquatic life cannot survive. The degradation of aquatic ecosystems not only affects biodiversity but also impacts communities that rely on these water.³⁸
- **Potassium (K):** While potassium is essential for plant health, its over-application through untreated piggery waste can lead to soil nutrient imbalances. Such imbalances can harm soil structure and fertility over time, making it more difficult for crops to thrive. Additionally, the environmental consequences of excessive potassium runoff can further exacerbate water quality issues.³⁹



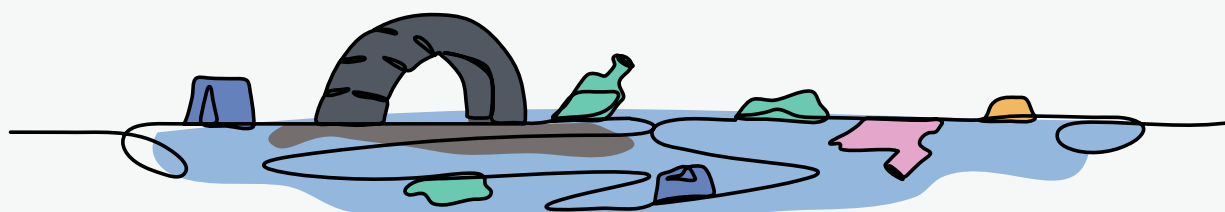
4.3. Environmental Pollution from Piggery Waste & Associated Risks

The improper management of piggery waste is a critical environmental issue that has garnered increasing attention due to its significant implications for water and soil quality. Piggery waste, if not managed appropriately, can lead to severe environmental pollution, affecting both human health and ecosystem integrity.

WATER POLLUTION:

One of the most pressing concerns associated with piggery waste is water pollution. Untreated or inadequately treated pig waste can contaminate nearby water bodies through runoff and leaching processes. The high levels of nitrogen compounds, particularly nitrates, are particularly concerning. When pig waste is applied to agricultural fields without proper management leads to run offs into water bodies causing eutrophication. This can deplete oxygen levels in water bodies, resulting in hypoxic conditions detrimental to aquatic life.⁴⁰

Open pools or lagoons used to store piggery wastewater pose a serious environmental threat. These pools act as reservoirs for a concentrated mix of nutrients, organic matter, pathogenic microorganisms, pharmaceutical residues, and antimicrobial-resistant (AMR) bacteria. Overflow, seepage, or deliberate spraying of lagoon contents can result in environmental contamination on a large scale.⁴¹ A significant number of piggery operations do not possess adequate wastewater treatment systems. As a result, untreated or poorly treated effluents are often discharged directly into the environment. This exacerbates the risk of groundwater and soil contamination, promotes the spread of AMR organisms, and undermines public health and environmental safety.⁴²



SOIL CONTAMINATION:

The indiscriminate application of untreated piggery waste on agricultural land is a major contributor to soil contamination. Pig manure, when not properly managed, introduces high loads of nutrients (particularly nitrogen and phosphorus) that disrupt soil chemistry, leading to acidification, nutrient imbalances, and a decline in soil fertility. These imbalances not only hinder plant growth but also undermine long-term agricultural productivity. Furthermore, the accumulation of heavy metals in soils due to repeated applications of pig manure raises concerns about food safety. Research indicates that heavy metal concentrations in crops grown on contaminated soils can exceed safe limits, posing potential health risks to consumers.⁴³ The presence of antimicrobial residues in untreated pig manure poses additional risks. These issues are explored in detail in the Section on AMR.

AIR TOXIFICATION:

Piggery units generate substantial quantities of waste that contribute significantly to air pollution, primarily through the emission of harmful gases such as ammonia, methane, hydrogen sulphide, and particulate matter.⁴⁴ Ammonia, emitted from pig waste, irritates the respiratory tract and reacts in the atmosphere to form PM_{2.5}, a harmful pollutant that travels deep into the lungs and circulates through the bloodstream.⁴⁵ Methane, produced via anaerobic fermentation of slurry, is a potent greenhouse gas, while hydrogen sulphide and dust particles further exacerbate air quality issues, leading to mucosal irritation, increased stress, and decreased quality of life for surrounding communities.⁴⁶

The health risks associated with air pollution from piggery waste are both acute and chronic. Chronic exposure to high concentrations of ammonia and dust, common by products of piggery operations, can lead to respiratory diseases such as bronchitis and asthma, while exposure to microbial by-products and pathogens present in pig waste increases the risk of infections and other health complications.⁴⁷

Cardiovascular risks are another serious concern. There may be an increased risk of cardiovascular diseases associated with exposure to air pollution. Studies accessing the general impact of air pollution suggest that long-term exposure to PM_{2.5} may accelerate adverse effects on cardiovascular health. The risk may be considerably higher among the elderly, children, and individuals with pre-existing cardiovascular and respiratory conditions.⁴⁸

Moreover, poor air quality has been associated with heightened vulnerability to respiratory infections, including viral diseases such as swine flu (H1N1).⁴⁹ Studies suggest that polluted environments facilitate both the transmission and severity of respiratory infections.⁵⁰ Symptoms such as fever, cough, and respiratory distress tend to worsen in high-pollution zones, placing disproportionate strain on nearby healthcare systems.

PATHOGEN TRANSMISSION:

Piggery waste contains a diverse range of pathogenic bacteria, viruses, and parasites, which can present significant risks to both environmental and human health. Exposure can occur through direct contact with waste, consumption of contaminated crops, or ingestion of water polluted by piggery effluent. Of particular concern are waterborne disease outbreaks linked to the contamination of drinking water and recreational sources by pig waste. In regions lacking effective waste treatment systems, such contamination can lead to rapid transmission of illnesses such as gastroenteritis, dysentery, typhoid, malaria and other serious infections.⁵¹ Persistent contamination of water bodies also degrades aquatic ecosystems and places long-term public health at risk.⁵²

BIOMEDICAL WASTE AND CARCASS DISPOSAL:

In addition to the substantial amounts of manure produced by piggery operations, these facilities also generate biomedical waste, including animal carcasses. When improperly managed, such as through the use of uncovered disposal pits or the pooling of stormwater, these waste products can create significant public health risks. Decomposing carcasses serve as breeding grounds for pathogens and attract vectors such as rodents and flies, facilitating disease spread both on-site and in neighboring areas.⁵³ Contaminants from improperly disposed carcasses can also leach into surface and groundwater, further compounding risks of waterborne infections and environmental degradation.⁵⁴



ANTIMICROBIAL RESISTANCE (AMR) RISKS:

The emergence and proliferation of AMR in piggery waste is a critical concern for both environmental and public health. Piggery operators have been observed to be routinely using antibiotics for disease prevention, growth promotion, in addition to therapeutic purposes. This widespread use exerts selective pressure on microbial communities, leading to the enrichment of antibiotic-resistant bacteria (ARB) and the accumulation of antibiotic resistance genes (ARGs) in pig manure, wastewater, and the surrounding environment.⁵⁵

Metagenomic studies reveal that piggery environments contain a diverse resistome, with ARGs conferring resistance to at least 18 classes of antibiotics, such as tetracyclines, sulfonamides, quinolones, aminoglycosides, and multidrug efflux systems. These genes are not restricted to pigs' gut microbiota, they are widely dispersed across the infrastructure, effluent, nearby soil, and slaughterhouse settings, indicating extensive environmental spread.⁵⁶

AMR determinants can persist in the environment for extended periods, even after the cessation of antibiotic use, due to the stability of genetic material and the potential for ongoing gene transfer among environmental bacteria.⁵⁷ As a result, piggery operations may act as long-term reservoirs for resistance traits, increasing the likelihood of their transfer to clinically significant pathogens and complicating efforts to combat antibiotic resistance on a broader scale.⁵⁸

Conclusion and Recommendations

5.1. Conclusion

This study concludes that piggery operations in India pose severe and escalating risks to public health and environment integrity, stemming from the unchecked proliferation of units operating with limited regulatory framework. The indiscriminate disposal of manure, slurry, carcasses, and biomedical waste contributes to significant air, water, and soil pollution, manifesting in the contamination of surface and groundwater sources and the unchecked release of hazardous gases such as methane, ammonia and

hydrogen sulphide in the air creating pathways for respiratory illness, gastrointestinal infections, vector-borne diseases, and antimicrobial resistance (AMR), which the World Health Organization has termed a “silent pandemic”.

Field findings from 146 units across the country and scientific literature reviewed in this paper point to an industry largely defined by unregulated operations, where poor infrastructure and an absence of technical oversight create conditions for environmental degradation and potential public health hazard.

Addressing these risks requires more than incremental change. A robust, multi-stakeholder response, grounded in statutory obligations, is urgently needed.

Ultimately, the continued tolerance of unregulated piggery operations permits widespread environmental degradation and endangers vulnerable human populations, including children and elderly living in proximity who face heightened risk of respiratory illness, enteric infections, and vector-borne diseases, under the pretext of economic activity. Only through coordinated environmental and epidemiological intervention can the hidden health externalities of this sector be contained, ensuring cleaner air, safer water, and dignified living conditions for all.

5.2. Specific Policy Recommendations

The following recommendations emphasize the need for improved waste management policies. By implementing these strategies, stakeholders can work collaboratively to minimize environmental impacts, safeguard public health, and promote the long-term sustainability of piggery units across India.

A. Central Government - Ministry of Environment, Forest and Climate Change

i. Include Piggery Sector in Environmental Impact Assessment (EIA)

Regulations: Bring large-scale piggeries under the purview of the EIA Notification under the EPA, 1986, given their proven impact on air, water, and soil quality.

ii. Promote Composting and Biochar Production: Encourage piggery units to adopt composting and biochar technologies for on-site waste treatment, reducing pathogen loads and producing organic soil amendments.

B. Central Pollution Control Board (CPCB)

i. Urgent Reclassification of Piggeries: The current Pollution Index (PI) assigned to piggery units (PI: 50) is dangerously misleading. At present, the piggery industry has not been scored for wastewater and biomedical waste. As such, an accurate assessment would very likely lead to higher PI for piggery units. CPCB must carry out an assessment after an accurate on ground sample and revise the classification accordingly.

ii. Develop and Notify Comprehensive Environmental Guidelines for Piggeries: Drawing from existing guidelines for poultry and dairy units, CPCB must create enforceable standards or guidelines on Consent to Establish (CTE) and Consent to Operate (CTO) requirements and procedures. Additionally, effluent treatment, solid waste handling, odour control mechanisms and measures for bioaerosol and particulate matter containment should be detailed in Guidelines of this nature.

iii. Mandate Environmental Monitoring and Reporting: Requirement should be placed on all piggery units to maintain and publicly disclose logs on waste generation, treatment, carcass disposal, and antibiotic/hormone usage.

iv. Integrate Nutrient Recovery and Solid-Liquid Separation Systems: Mandate or incentivize the use of nutrient recovery technologies and solidliquid separators for large and medium-scale units to reduce water pollution and enable the transport of stabilized manure solids.

C. State Pollution Control Boards (SPCBs)

i. Strict Enforcement of Consent Mechanisms: No piggery unit should be allowed to operate without a valid CTE/CTO. SPCBs must routinely inspect piggery units and impose penalties or closures for non-compliance.

ii. Monitor and Regulate Air and Water Emissions: Enforce statutory standards on ammonia, methane, and hydrogen sulphide emissions; and ensure wastewater treatment complies with norms under the Water Act.

iii. Public Grievance Mechanism: Establish online platforms where communities affected by nearby piggeries can lodge complaints regarding foul odour, fly infestation, or contamination.

iv. Implement Guidelines for Scientific Disposal of Food Waste: Ensure implementation of Mechanism/Guidelines for Control of Pollution and Enforcement of Environment Norms at Individual Establishments and the Area/ Cluster of Restaurants/ Hotels/ Motels/ Banquets etc, 2020 to ensure that the use of leftover food waste as pig feed is curbed.

v. Integrate Internet of Things and Real-Time Monitoring: Encourage the adoption of low-cost IoT systems to monitor waste storage and discharge, enabling real-time compliance checks.

D. Urban Local Bodies and State Animal Husbandry Departments

i. Zoning and Land Use Controls: Establish and enforce siting restrictions for piggery units to ensure they are not located within 200 meters of residential areas, schools, hospitals, or water sources.

ii. Mandatory Sanitary Infrastructure: Requirement to be placed on all piggery units to install closed drainage systems, use waste segregation bins (biodegradable, biomedical, and carcass waste) and ensure covered storage for feed and carcasses to control vector breeding.

iii. Facilitate Composting and Treated Water Reuse: Support units to set up composting areas and assess the reuse of treated wastewater for irrigation or as liquid fertilizer inputs.



E. Piggery Operators and Industry Associations

i. Adopt Circular Economy Models: Invest in bio-digesters, microalgae-based wastewater treatment (as per Nagarajan et al., 2019), and composting systems to minimize ecological harm and recover energy or manure from waste.

ii. Discontinue Hazardous Practices: Stop the use of non-therapeutic antibiotics and heavy metals such as copper and zinc in feed, which have been found to contaminate the food chain and degrade soil health.

iii. Humane and Scientific Carcass Disposal: Implement lime burial or approved incineration technologies. Carcasses must never be dumped in open drains or fed to dogs, which violates both public health standards and basic decency.

iv. Health and Safety Compliance: Equip workers with PPE, conduct periodic health check-ups, and provide training on sanitation, disease prevention, and humane handling of animals.

F. Veterinary and Public Health Authorities

Zoonotic Disease Surveillance: Monitor and report incidences of zoonotic infections linked to piggery waste and bioaerosols, with focus particularly on swine-origin pathogens such as Salmonella, E. coli, and Listeria.

ii. Combat Antimicrobial Resistance (AMR): Regulate antibiotic usage in piggery operations. Routine feed additives must not be utilised unless prescribed by a registered veterinarian.

iii. Test Groundwater and Soil Near Piggeries: Establish regular testing regimes for groundwater and soil quality, especially where carcass pits or slurry ponds are used.



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